# Part A: Theory (Short Questions)

## 1. Explain Mean, Median, and Mode with a real-life example.

**The mean is the mathematical average of a dataset, calculated by summing all values and dividing by the total count. The median represents the middle value when data is sorted in ascending order, making it resilient to extreme outliers. The mode is the most frequently occurring value in the dataset. For instance, consider a small shoe store tracking daily sales sizes: $7,8,8,8,9,$ and $12$. Here, the mean size is $8.6$, the median is $8$, and the mode is $8$, illustrating how each metric summarizes the distribution differently.**

## 2. What is the difference between Variance and Standard Deviation?

**Variance measures the average squared deviation of each data point from the mean, showing how spread out the data is in squared units. Standard deviation is simply the square root of the variance, which brings the measure of dispersion back to the original units of the data. Because standard deviation uses the original units, it is much easier to interpret in real-world contexts, such as analyzing employee salaries or test scores.**

## 3. Define Normal Distribution and give one practical use case.

**A Normal Distribution is a continuous probability distribution characterized by a symmetric, bell-shaped curve where the mean, median, and mode all coincide at the center. It is defined by its mean and standard deviation, with most data clustering near the center and tapering off symmetrically toward the tails. A classic practical use case is modeling human heights or weights in a large population, where the vast majority of individuals cluster around the average, and very tall or very short individuals are increasingly rare.**

## 4. Explain Skewness and Kurtosis in simple words.

**Skewness measures the asymmetry of a probability distribution, indicating whether the data leans more heavily toward the left (negative skew) or the right (positive skew) relative to a normal bell curve. Kurtosis, on the other hand, measures the "tailedness" of a distribution, describing whether the tails are thin (light-tailed) or excessively fat with extreme outliers (heavy-tailed). Together, these metrics help analysts understand how much a dataset departs from a standard normal distribution.**

## 5. What is Probability? Differentiate between Empirical vs Theoretical Probability with examples.

**Probability is a numerical measure ranging from $0$ to $1$ that quantifies the likelihood of a specific event occurring. Theoretical probability is calculated based on known, logical possibilities under ideal conditions, such as determining that the chance of rolling a $3$ on a fair six-sided die is $1 / 6$. In contrast, empirical probability is derived from actual historical observations or experiments, such as rolling that same die $100$ times and finding it landed on $3$ exactly $18$ times ($18 / 100$).**

## 6. Explain Independent vs Dependent Events with one example each.

**Independent events are those where the outcome of the first event has zero influence on the probability of the second event occurring. For example, flipping a coin and getting heads does not change the odds of**

**getting heads on a second flip. Conversely, dependent events are those where the outcome of the first event directly alters the probability of the subsequent event. An example is drawing two cards from a standard deck without replacement, where drawing an Ace first decreases the total number of cards and changes the probability of drawing an Ace on the second draw.**

## 7. What is the intuition of Bayes Theorem in daily life?

**Bayes Theorem provides a mathematical framework for updating our beliefs or prior assumptions when new evidence becomes available. In daily life, it mirrors how humans naturally adjust their confidence levels; for instance, if you see dark clouds outside (new evidence), you update your initial belief that it was going to be a sunny day and decide to carry an umbrella. It allows us to continuously refine our predictions by combining prior knowledge with observed data.**

## 8. Explain Eigenvalue and Eigenvector in simple terms.

**An eigenvector is a special direction or vector in a matrix transformation that does not change its orientation or line of action, but only undergoes a scaling factor. The eigenvalue is that exact scaling factor by which the eigenvector is stretched, compressed, or flipped during the transformation. In data science applications like Principal Component Analysis (PCA), eigenvectors represent the principal axes of maximum variance, while eigenvalues quantify how much variance is captured along each of those axes.**